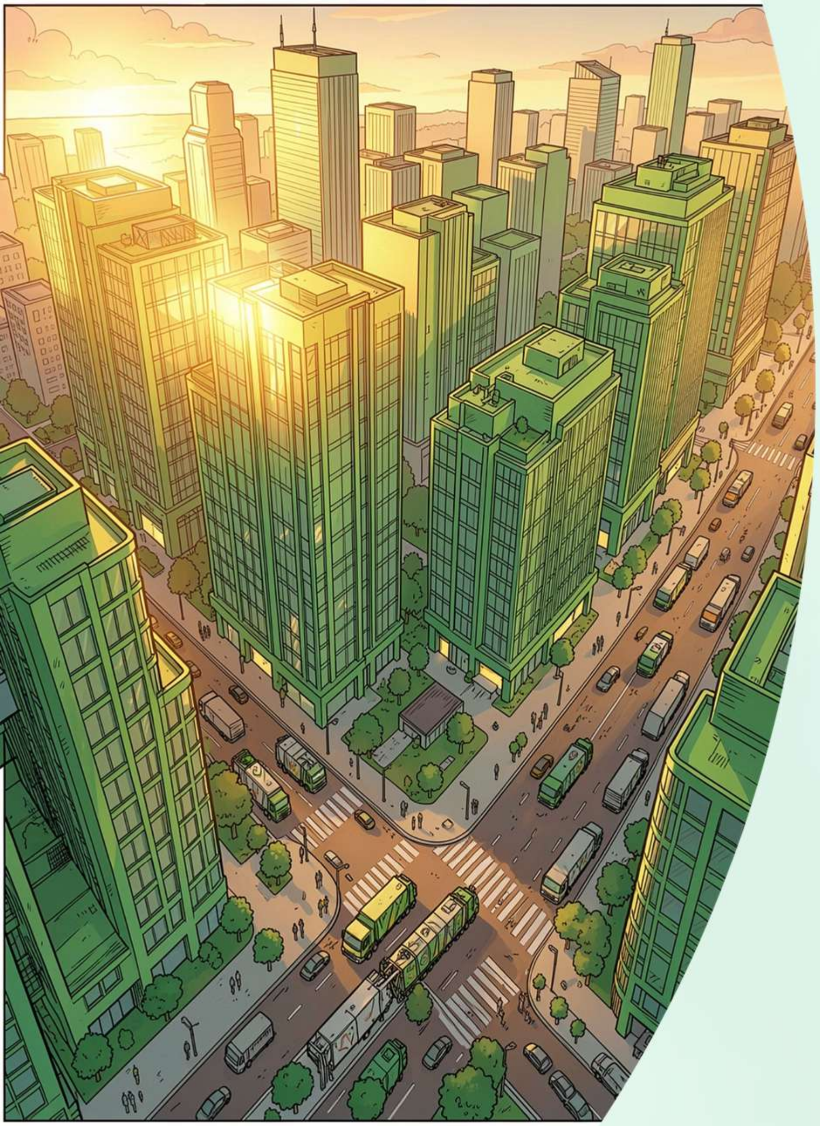


AI FOR SUSTAINABILITY ACADEMIC PROJECT

Municipal Waste Management RAG Assistant

Leveraging IBM Granite, retrieval-augmented generation, and artificial intelligence to transform how citizens, municipalities, and urban planners interact with complex waste management regulations—building smarter, cleaner, and more sustainable communities.

The Challenge

Citizens Struggle with Waste Disposal Information

Across rapidly urbanising cities, residents face a daily dilemma: *"Which bin does this go in?"* Municipal waste management guidelines are often buried across dozens of lengthy PDF documents, government portals, and fragmented policy frameworks. E-waste rules, plastic waste management amendments, and wet/dry segregation guidelines are each published separately by different regulatory bodies — making it nearly impossible for the average citizen to find accurate, actionable disposal information quickly.

This information gap leads to improper segregation at the source, contaminated recycling streams, and increased operational burdens on already strained municipal collection systems. The result is a measurable degradation of urban environmental quality and a direct setback to national sustainability targets.

SOLUTION

Introducing the RAG Assistant

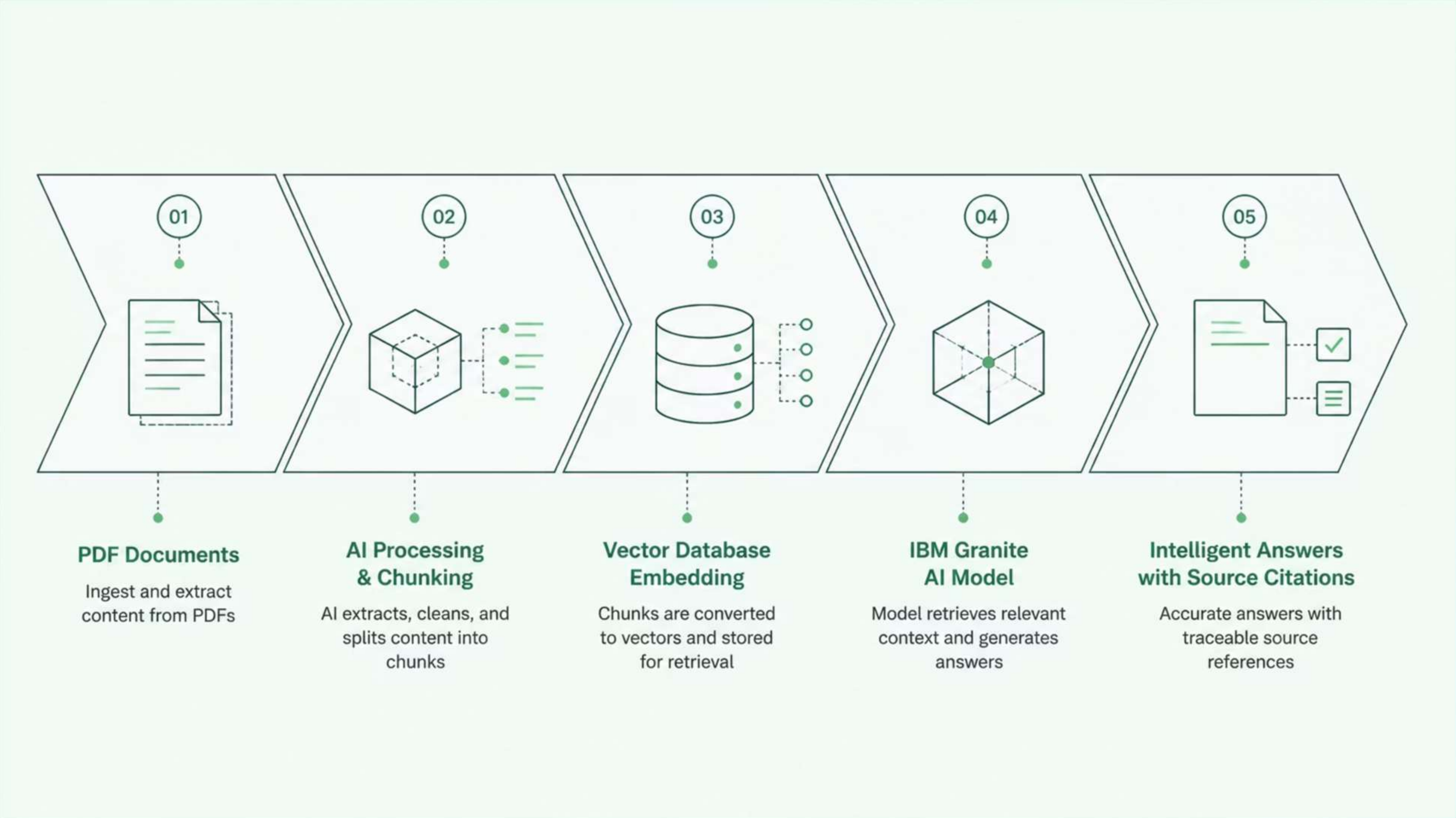
An AI-Powered Chatbot for Waste Management

The Municipal Waste Management RAG Assistant bridges the information gap using a Retrieval-Augmented Generation architecture. Instead of relying on a generic language model's pretrained knowledge, the system dynamically retrieves relevant passages from a curated corpus of official municipal and national waste management documents — including the E-Waste Management Rules, Plastic Waste Management Rules, and municipal segregation guidelines — and feeds them into the IBM Granite AI model to generate precise, source-cited answers.

Users simply ask natural-language questions through a clean, accessible dashboard interface. The assistant responds in seconds with accurate guidance and direct citations to the source regulatory documents, ensuring transparency and trust.

How Retrieval-Augmented Generation Works

The RAG pipeline transforms static government documents into an interactive knowledge base. Each stage of the process is designed to ensure that the AI's responses are grounded in authoritative text — not hallucinated — making it suitable for regulatory compliance and public information use.



By grounding every response in retrieved excerpts from verified government publications, the system eliminates hallucination risks and gives citizens confidence that the disposal guidance they receive is legally accurate and jurisdictionally correct.

Powering the System

IBM Granite: Enterprise-Grade AI for Public Good

At the core of the assistant lies **IBM Granite**, a family of large language models designed for enterprise and government deployments. Granite models are optimised for accuracy, transparency, and regulatory compliance — critical requirements when the outputs inform citizens about environmentally sensitive waste handling procedures.

Document Understanding

Granite processes complex regulatory language, tables, and structured policy text with high comprehension fidelity across long-form municipal documents.

Grounded Generation

Responses are generated strictly from retrieved context windows, ensuring factual alignment with source rules rather than relying on parametric memory.

Source Citations

Every answer includes explicit references to the originating document section, page, and rule number — building public trust through auditability.

Real Questions, Accurate Answers

Citizens interact with the assistant using natural, conversational language. Below are representative queries the system handles — each routed through the RAG pipeline and answered with citations from official Indian government waste management rules and municipal by-laws.

"How should I dispose of e-waste?"

The assistant retrieves the E-Waste Management Rules, 2022 and explains that residents must deposit old electronics at authorised collection centres or return them to manufacturers under Extended Producer Responsibility. It cites the specific rule and section.

"What is wet and dry waste segregation?"

Referencing the Solid Waste Management Rules, 2016, the system clarifies that wet waste is biodegradable kitchen waste while dry waste includes recyclables like paper, plastic, glass, and metal — and explains the two-bin system mandated by municipalities.

"How can plastic waste be recycled?"

Drawing from the Plastic Waste Management Rules, the assistant outlines the categories of recyclable plastics, the role of authorised recyclers, and the ban on single-use plastic items — with exact policy citations.

Sustainability Alignment

Advancing the UN Sustainable Development Goals

The Municipal Waste Management RAG Assistant directly contributes to two critical UN Sustainable Development Goals. By improving citizen access to accurate waste disposal information and reducing contamination in recycling streams, the project advances both urban sustainability and responsible consumption patterns at the municipal level.

SDG 11 — Sustainable Cities & Communities

Proper waste segregation at source reduces the environmental burden on urban infrastructure. By making disposal guidelines instantly accessible, the assistant fosters cleaner neighbourhoods, reduces landfill overflow, and strengthens municipal waste governance — key indicators under SDG Target 11.6.

SDG 12 — Responsible Consumption & Production

Informed citizens make better disposal decisions. By guiding residents to recycle plastics correctly, route e-waste through authorised channels, and segregate organic waste for composting, the assistant promotes circular economy practices and supports SDG Target 12.5 on reducing waste generation through prevention, reduction, recycling, and reuse.

System Dashboard

A Clean, Accessible Interface for Every Citizen

The assistant's front-end is designed for inclusivity. A minimalist chat interface — rendered in a green-themed sustainability palette — allows users of all technical proficiencies to ask questions in plain language. Behind the scenes, a monitoring dashboard tracks query categories, response accuracy, and document coverage, giving municipal administrators real-time insight into citizen information needs.

Conversational Interface

Natural language input with no training required — citizens simply type or speak their waste question.

Source Transparency

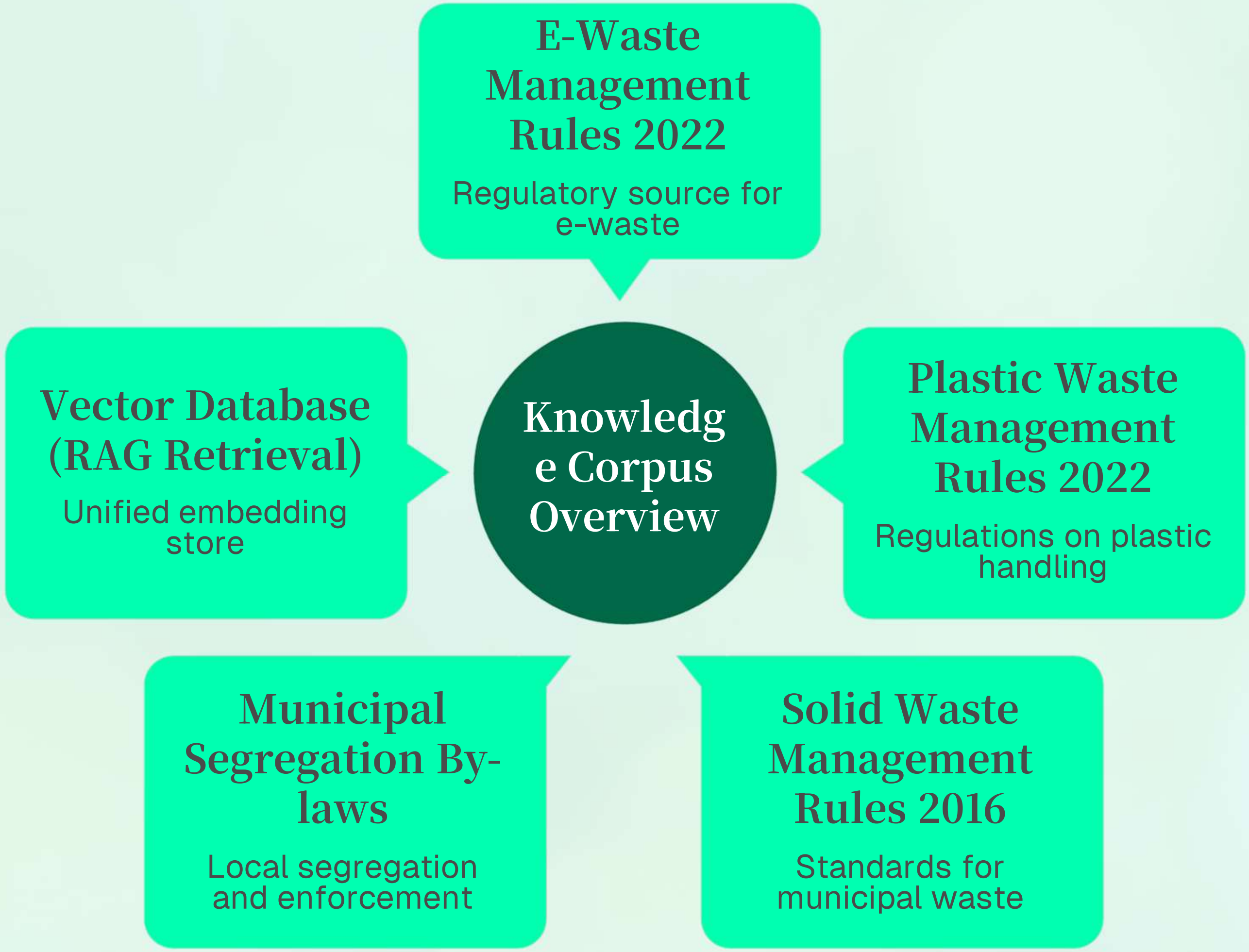
Every response links back to the exact PDF, section, and page number for full auditability.

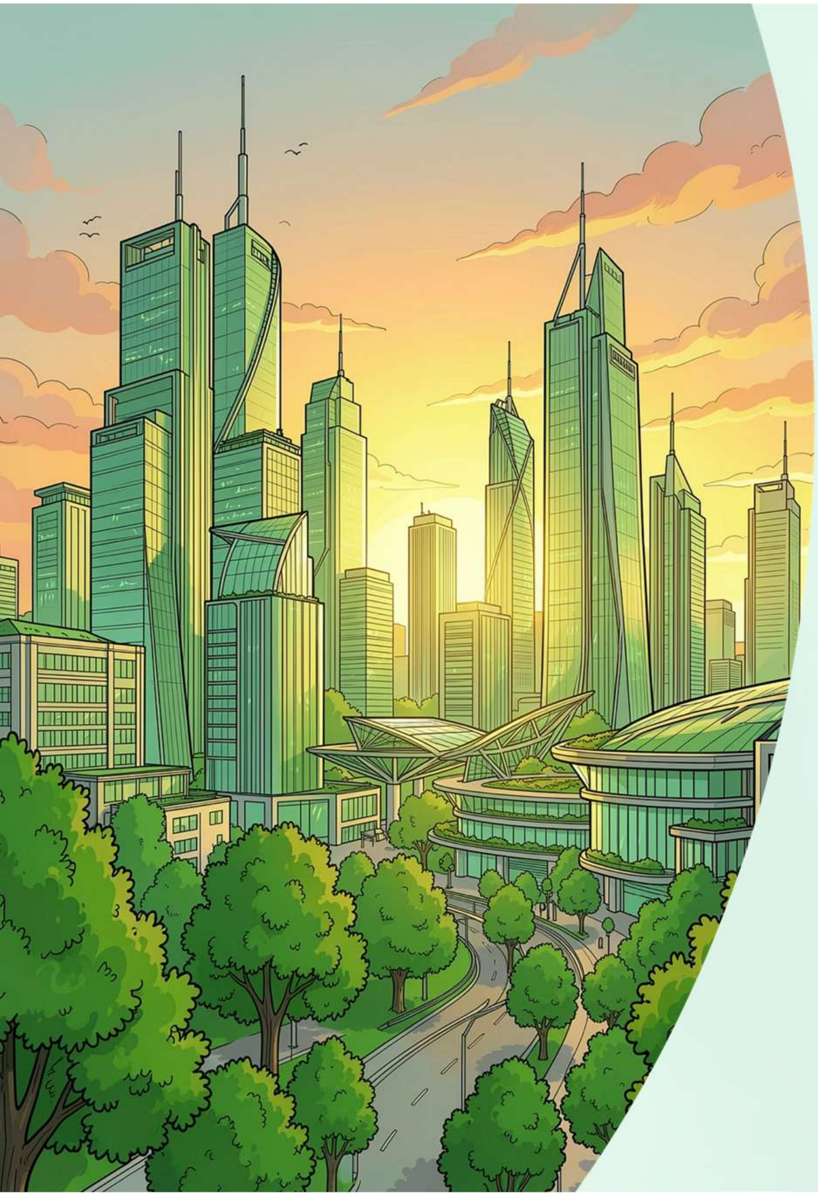
Admin Analytics

Municipal teams view aggregated query trends to identify gaps in public awareness and outreach.

Knowledge Corpus at a Glance

The RAG assistant draws from a carefully curated and continuously updated document library — each source selected for its regulatory authority and relevance to Indian municipal waste management.





Municipal Waste Management RAG Assistant

Smarter. Cleaner. Sustainable.

Leveraging IBM Granite, Retrieval-Augmented Generation, and AI for sustainable waste management — building smarter, cleaner, and more sustainable communities, one informed citizen at a time.

Project Impact

Bridging the gap between complex environmental regulation and everyday civic action through transparent, grounded, and accessible AI.

Next Steps

- Expand corpus to include state-level waste policies
- Pilot deployment with select municipal corporations
- Multilingual support for Hindi and regional languages
- Integration with municipal mobile applications